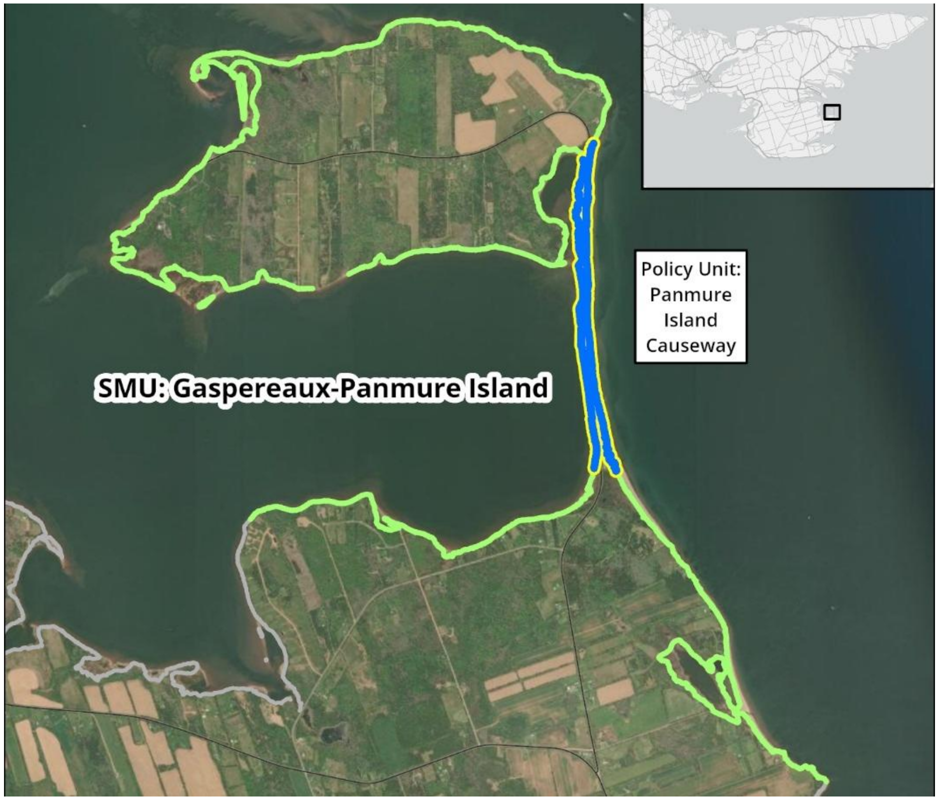
DRAFT

Gaspereaux-Panmure Island

Shoreline length: 28 km



Causeway and Provincial Park to the right

Armouring along cliff updrift of Panmure Island Provincial Park.

Bluff at Panmure Island Lighthouse

Shore Types

This SMU includes Gaspereaux shores, Panmure Island and its causeway. Main shore types are dunes, low plains, wetlands, and cliffs.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
39	5	4	13	19	0	20	0

13-Aug-26

Eroding bluff feeding a beach on north shore of Panmure Island

Governance, Infrastructure, and Economy

Gaspereaux-Panmure Causeway is entirely within the Rural Municipality of Three Rivers. The causeway was built on a barrier dune and provides the sole access route to Panmure Island and is managed by PEI DTI who have conducted local coastal engineering maintenance work since 2005. Infrastructure includes Reilly's Road (Gaspereaux), Panmure Island Road (Route 347, Gaspereaux), the Panmure Head Lighthouse, and Route 347 (Chemin Panmure Island). Local and legacy land use constraints include several small lots and several unpaved roads, including the causeway itself; the single access road is a flood hazard risk, a special consideration, since it is the only route on or off Panmure Island.

Natural Assets and Heritage

The area includes the Panmure Island Lighthouse, and Panmure Island Cultural Grounds, significant as the site of the annual Abegweit Pow Wow. Panmure Island Provincial Park beach is located at the south end of the causeway. The shoreline includes critical habitat for bird species including the endangered Piping Plover and threatened Bank Swallow.

Planning and Policy Framework

Three Rivers has municipal planning authority. Development on the coast is informed by the Environmental Protection Act, including a 15m buffer, with no additional building setbacks aside from the buffer. The Official Plan includes specific guidance permitting armouring, though such provisions are secondary to any provincial standards, and includes some policies regarding expansion of buildings within the sea level rise overlay. There are no specific flood elevations or vertical setbacks established, but the Official Plan does include specific erosion-based setbacks, with the possibility of reduced setbacks under certain conditions. Hazards are assessed at the subdivision stage.

Coastal Flood and Erosion Risk

The causeway is at risk of flooding throughout, and under erosion pressure along its north half. The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential shoreline change to future time horizons.

Number of at-risk buildings [N. per km shoreline] in SMU

	Present	0.3m SLR (2050)	0.6m SLR (2080)	1.2m SLR (2130)
Flood risk	11 [<1/km]	15 [1/km]	22 [1/km]	31 [1/km]
Erosion risk	N/A	11 [<1/km]	26 [1/km]	48 [2/km]

Sediment transport – Net sediment transport is high along the outer coast, where it converges toward the Provincial Park beach at the south end of the causeway. It is moderate around Panmure Islands, where it feeds beaches at Billhook Point and Marsh Point.

Proposed SMP Strategies

It is essential that sediment supply is maintained to the causeway beach, both to preserve the recreational value of the Provincial Park beach, as well as the storm protection value for the causeway link itself. This would be achieved by a strategy of NAI updrift of the causeway, followed by MR where assets are at risk including but not limited to the Lighthouse at Panmure Head.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
Proposed SMU strategy	**NAI**	**MR**	**MR**
Rationale	Mid- to long-term MR will reduce risk to the limited number of vulnerable assets, allowing continued sand supply to the causeway beach.		
Proposed PU strategy **Panmure Island Causeway**	**R**	**R**	**R**
Rationale	The causeway is a critical link to the Panmure Island community. Resist strategy must incorporate sand nourishment to sustain Provincial Beach Park.		

LEGEND

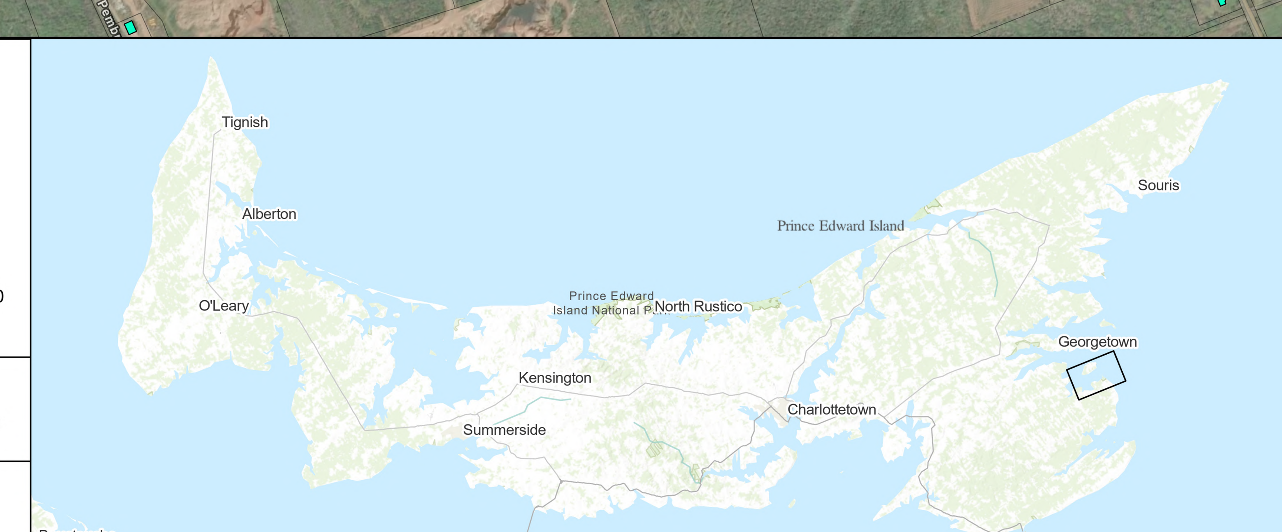
- Policy Unit
- Shoreline Management Unit
- Net Longshore Sediment Transport
- High Hazard Area
- Flood Extent - Present Day (100-year Return)
- Flood Extent - 1.2m SLR (100-year Return)
- Property Boundary
- Building > 90ft²

N
W E
S

0 130 260 520
Metres

Coordinate System:
NAD 83 CSRS PEI Double Stereographic

NOTES:



COASTAL EROSION AND FLOOD HAZARD

Shoreline Management Unit: Gaspereaux-Panmure Island

Prince Edward Island
Île-du-Prince-Édouard

CBCL